

# Final Exam

Name: \_\_\_\_\_

The exam contains 7 pages (including this cover page) and 6 questions. The total number of possible points is 50. You have 120 minutes to complete the exam.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **No outside resources allowed.** You may not use any notes, calculators, textbooks, or other resources on this exam. Make sure all electronic devices are silenced and put away.
- **Partial credit will be given**, so write down as much as you know even if some part of the proof might be missing.
- **Read the questions carefully**, make sure you answer the question being asked and understand the statement of the problem.
- **Answer all questions in the space provided**, you may use the back of pages as scrap paper, but only what is written under the question will be graded.

| Question | Points | Score |
|----------|--------|-------|
| 1        | 10     |       |
| 2        | 12     |       |
| 3        | 7      |       |
| 4        | 7      |       |
| 5        | 7      |       |
| 6        | 7      |       |
| Total:   | 50     |       |

Do not write in the table to the right.

1. (10 points) Determine if each of the following statements is TRUE or FALSE by completely filling in the corresponding bubble. To receive full credit you must provide a short explanation or sketch of a proof.

**You do not need to provide a formal proof.**

- (a) Every bounded function that is bounded on a closed interval  $[a, b]$  attains a maximum and minimum value on  $[a, b]$ .

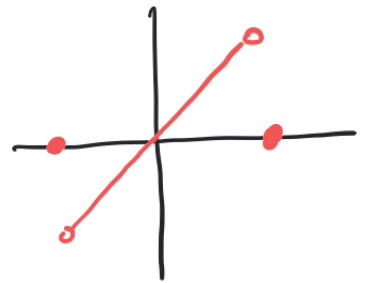
☐ TRUE

☒ FALSE

EVT requires continuity.

Counterexample:

$$f(x) = \begin{cases} x & x \in (-1, 1) \\ 0 & x = -1, 1 \end{cases}$$



- (b) Every function that attains maximum and minimum value on a closed interval  $[a, b]$  is bounded on  $[a, b]$ .

☒ TRUE

☐ FALSE

Take  $B = \max\{|M|, |m|\}$

then  $|f(x)| \leq B$  for all  $x \in [a, b]$

- (c) If  $g$  is defined on  $[a, b]$  and  $g = f'$  for some differentiable function  $f$ , then  $g$  is continuous on  $[a, b]$ .

☐ TRUE

☒ FALSE

there exist discontinuous derivatives.

- (d) If  $f$  is differentiable on  $[a, b]$ , then  $f$  is integrable on  $[a, b]$ .

☒ TRUE

☐ FALSE

differentiable  $\Rightarrow$  continuous  $\Rightarrow$  integrable  
on  $[a, b]$  on  $[a, b]$  on  $[a, b]$

- (e) Every continuous function on  $(a, b)$  is integrable.

☐ TRUE

☒ FALSE

Integrability was only defined for  
bounded functions on closed  
intervals.

2. (12 points) For each of the following, give an example satisfying the given conditions OR give a brief explanation why such an example does not exist.

- You do *not* need to prove your example satisfies the conditions.
- To receive full credit **you must give an algebraic definition**, a sketch will earn you partial credit.

(a) A sequence that is bounded but does not converge.

$$s_n = (-1)^n$$

(b) A real-valued function  $f$  defined on  $\mathbb{R}$  such that  $f$  is uniformly continuous but  $f^2$  is not uniformly continuous.

$$f(x) = x$$

(c) A real-valued function  $f$  defined on  $\mathbb{R}$  such that every lower sum equals every upper sum.

$$f(x) = 7$$

(d) A real-valued function  $f$  defined on  $[0, 1]$  that is *not* integrable but  $|f|$  is integrable.

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \notin \mathbb{Q} \end{cases}$$

3. (7 points) Prove that for all natural numbers  $n \in \mathbb{N}$ , if  $f(x) = x^2 e^x$ , then

$$f^{(n)}(x) = (x^2 + 2nx + n(n-1))e^x,$$

where  $f^{(n)}(x)$  is the  $n$ th derivative of  $f(x)$ . You may use without proof the fact that if  $g(x) = e^x$ , then  $g'(x) = e^x$ .

Prove by induction.

Base case:  $f(x) = x^2 e^x$        $f'(x) = x^2 e^x + e^x \cdot 2x$   
 $= (x^2 + 2x) \cdot e^x \checkmark$

Inductive Step: Assume  $f^{(n)}(x) = (x^2 + 2nx + n(n-1))e^x$

$$\begin{aligned} f^{(n+1)}(x) &= \left( f^{(n)}(x) \right)' = (x^2 + 2nx + n(n-1))e^x + \\ &\quad e^x (2x + 2n) \\ &= (x^2 + 2(n+1)x + (n+1)n)e^x \checkmark \end{aligned}$$

Therefore by the principle of mathematical

induction,  $f^{(n)}(x) = (x^2 + 2nx + n(n-1))e^x$  for all

$n \in \mathbb{N}$ .

4. (7 points) Prove that the function

$$f(x) = \begin{cases} 1 & 0 \leq x < 1 \\ 5 & x = 1 \\ 2 & 1 < x \leq 2 \end{cases}$$

is integrable and find  $\int_0^2 f(x) dx$ .

$f$  is bounded by 5.  
Let  $\epsilon > 0$ . Let  $P = \{0, 1 - \frac{\epsilon}{10}, 1 + \frac{\epsilon}{10}, 2\}$ . Then

$$\begin{aligned} U(f, P) &= 1 \left(1 - \frac{\epsilon}{10} - 0\right) + 5 \left(1 + \frac{\epsilon}{10} - 1 + \frac{\epsilon}{10}\right) + 2 \left(2 - 1 + \frac{\epsilon}{10}\right) \\ &= 1 - \frac{\epsilon}{10} + 5 \left(\frac{\epsilon}{5}\right) + 2 \left(1 + \frac{\epsilon}{10}\right) \end{aligned}$$

$$\begin{aligned} L(f, P) &= 1 \left(1 - \frac{\epsilon}{10} - 0\right) + 1 \left(1 + \frac{\epsilon}{10} - 1 + \frac{\epsilon}{10}\right) + 2 \left(2 - 1 + \frac{\epsilon}{10}\right) \\ &= 1 - \frac{\epsilon}{10} + \frac{\epsilon}{5} + 2 \left(1 + \frac{\epsilon}{10}\right) \end{aligned}$$

$$U(f, P) - L(f, P) = \epsilon - \frac{\epsilon}{5} < \epsilon.$$

Therefore  $f$  is integrable.

$$\begin{aligned} \int_0^2 f(x) dx &= \int_0^1 f(x) dx + \int_1^1 f(x) dx + \int_1^2 f(x) dx \\ &= 1 + 0 + 2 = 3. \end{aligned}$$

5. (7 points) Let  $(s_n)$  be a sequence. Prove that if  $\lim s_n = \infty$  and if  $(t_n)$  is a bounded sequence, then  $\lim(s_n + t_n) = \infty$ .

let  $M > 0$  be given. Since  $t_n$  is bounded,

$\exists B$  s.t.  $|t_n| \leq B$  for all  $n$ . Since

$\lim s_n = \infty$ ,  $\exists N$  s.t. for all  $n > N$ ,

$s_n > M + B$ . Note that  $t_n \geq -|t_n| \geq -B$

so  $s_n + t_n \geq s_n - |t_n| > M + B - B = M$ .

Therefore  $\lim (s_n + t_n) = \infty$ .



6. (7 points) Let  $f$  be a function defined on  $[0, 1]$  that is continuous on  $[0, 1]$  and differentiable on  $(0, 1)$ . Suppose that  $\text{dom}(f) = [0, 1]$  and  $f(x) \in [0, 1]$  for all  $x \in [0, 1]$ . If  $f'(x) \neq -1$  for all  $x \in (0, 1)$ , prove that there is **exactly one** point  $x_0 \in [0, 1]$  such that  $f(x_0) = 1 - x_0$ .

we first show there exists at least one  $x_0 \in [0, 1]$  s.t.  $f(x_0) = 1 - x_0$ . If  $f(0) = 1$  or  $f(1) = 0$  we are done, so assume  $f(0) < 1$  and  $f(1) > 0$ . Let  $g(x) = f(x) - (1 - x)$ .  $g(x)$  is continuous on  $[0, 1]$  and diff. on  $(0, 1)$ .

Note that  $g(0) = f(0) - 1 < 0$  and

$g(1) = f(1) > 0$  so by the IVT

$\exists$  at least one point in  $(0, 1)$

so that  $g(x_0) = 0 \Rightarrow f(x_0) = 1 - x_0$ .

we will prove there exists exactly one point by contradiction. Assume  $x_1$  and

$x_2$  both have the property that

$f(x_1) = 1 - x_1$  and  $f(x_2) = 1 - x_2$ . Assume

WLOG  $x_1 < x_2$ .  $f$  is cont. on  $[x_1, x_2]$  and differentiable on  $(x_1, x_2)$  so by the

MVT  $\exists c$  s.t. 
$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{1 - x_2 - (1 - x_1)}{x_2 - x_1} = -1$$

A contradiction.